



Protocols for Co-Cognition

The operating layer between AI adoption and durable value

FORMAL PROTOCOL THEORY × PROTOCOLS FOR BUSINESS

Robert Peake · 2026



GOAL
criteria



GENERATE
candidates



JUDGE
adequacy



VERIFY
evidence



REPAIR
error



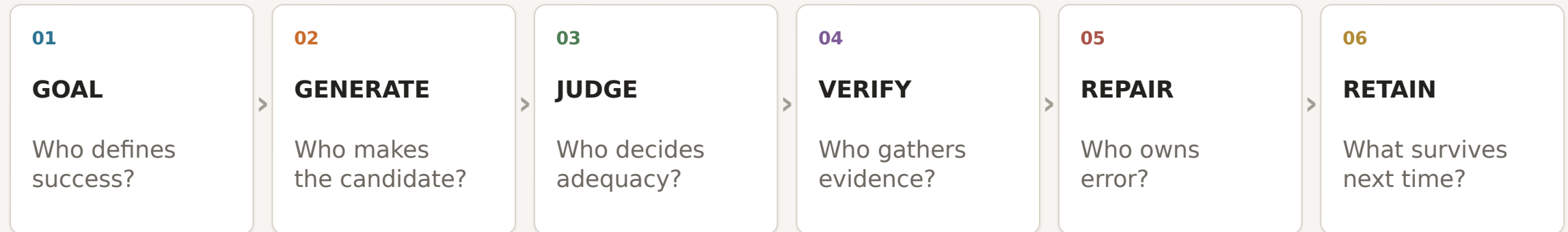
RETAIN
capability

protocol = conditional coordination

Every business already runs an AI protocol

When rules stay tacit, interface, incentive and habit write them for you.

USUALLY TACIT



A prompt governs a response. A protocol governs entry, roles, evidence, handoffs, exceptions, recourse and memory.

DURABLE ABOVE MODEL CHURN

Adoption is access. Protocolization is redesign.

The unit of value moves from tool presence to the division of cognitive labour.

ADOPTION DASHBOARD

Capability is present

- licences and active users
- tokens and task volume
- hours saved
- outputs shipped

PROTOCOLIZED WORK

Capability is organised

- decision ownership
- verification placement
- handoff quality and recourse
- retained expertise and recovery

CYCLE TIME

RECIPIENT REWORK

TRACEABILITY

RECOVERY

A unit, a state, and an operator

A compact formal object for studying coordination, fit and change.

INTERACTION REGIME

1

Unit of analysis

Recurring allocation of cognition and decision rights

POSTURE

2

State estimate

Trust × Sovereignty, with Load and Debt signals

PROTOCOL

3

Conditional operator

Rules that reorganise the regime and its trajectory

STATE TRANSITION

$$\mathbf{z}_{t+1} = \mathbf{F}(\mathbf{z}_t, \pi, \mathbf{x}_t, \mathbf{y}_t)$$

state protocol context feedback

PROTOCOL FIT

Which procedure works best from this state, for this task and these stakes?

PROTOCOL TRANSITION

How does repeated use change judgement, load, trust and future capability?

TODAY'S WORKFLOW TRAINS TOMORROW'S WORKFORCE

The procedure changes the outcome

Same capability, different interaction rules—and different trajectories.

BASTANI ET AL. · ~1,000 STUDENTS

Assisted practice

Later unaided exam

GPT BASE

+48%

−17% vs control

GPT TUTOR

+127%

≈ control

Interaction rules changed both assisted performance and what remained without assistance.

DELL'ACQUA ET AL. · 758 CONSULTANTS

The capability frontier is jagged

INSIDE

**speed +
quality
gains**

OUTSIDE

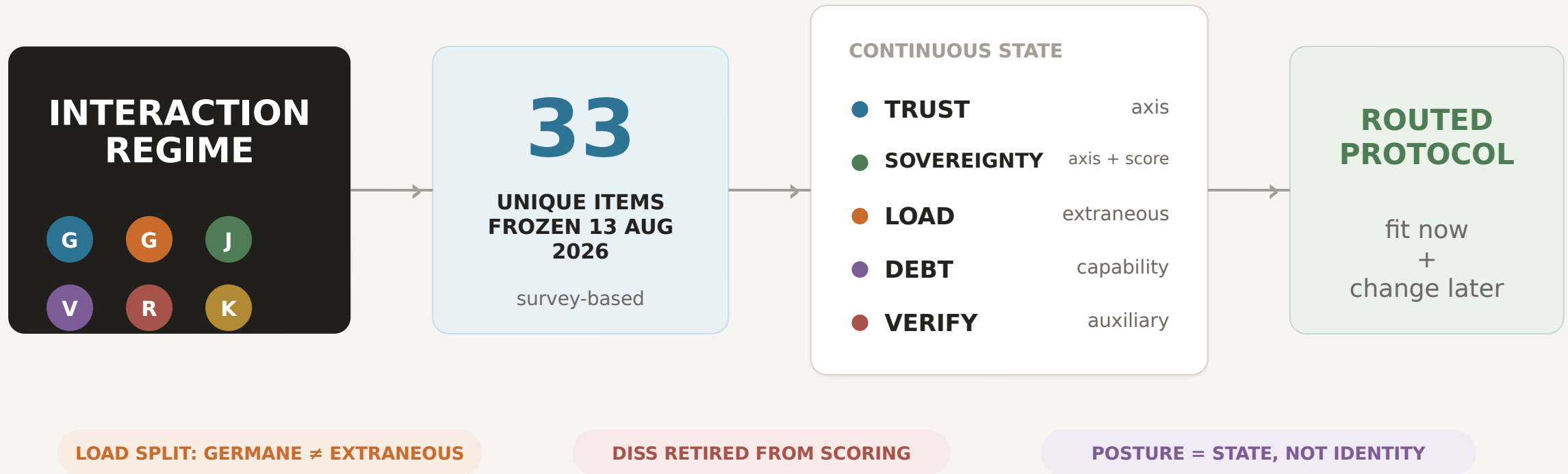
**correctness
fell**

The operational rule must be conditional on task and demonstrated capability.

PROTOCOL IS A CAUSAL OBJECT

The assessment makes the hidden regime observable

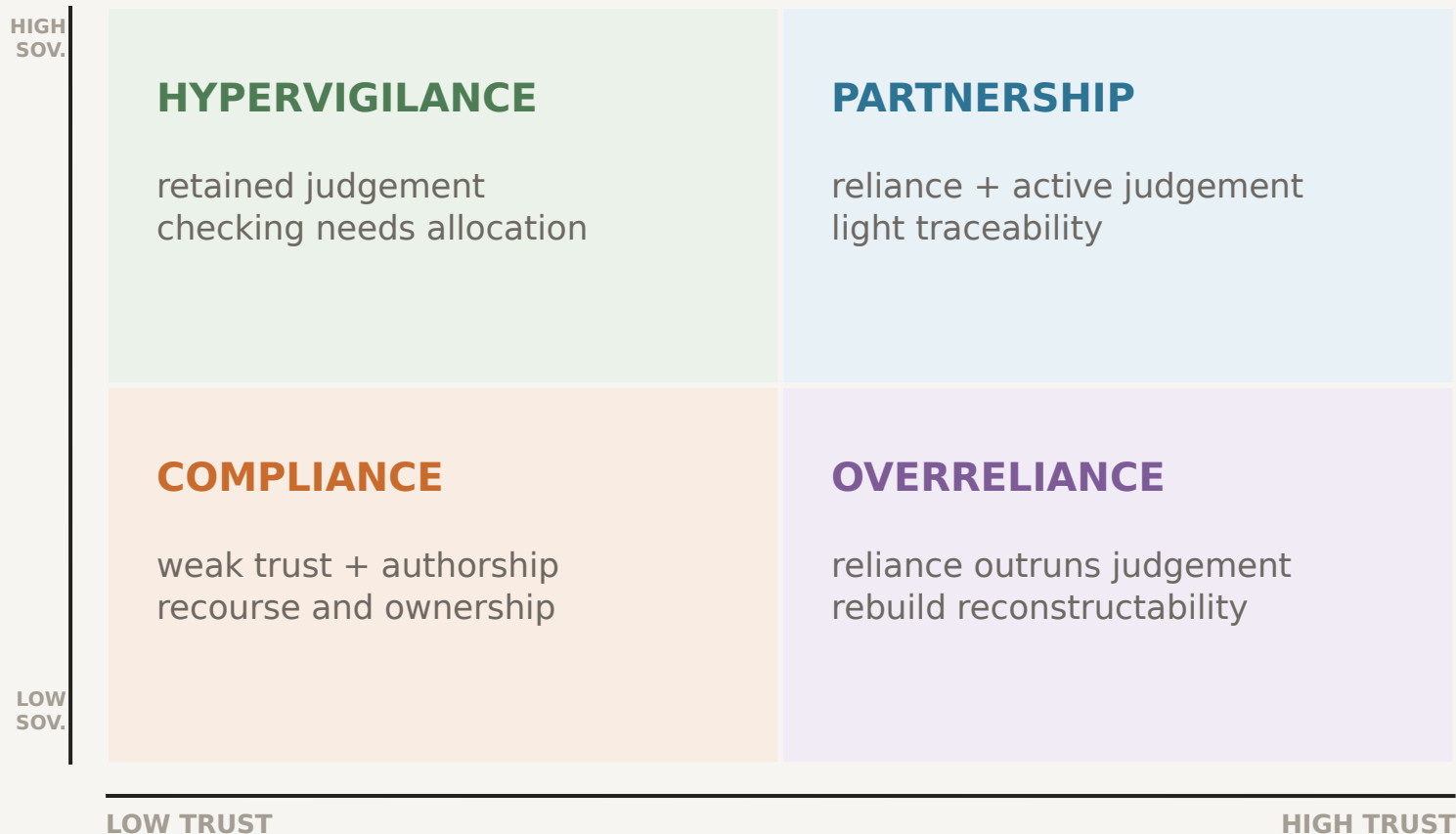
A 33-item state estimator connects lived interaction to testable protocol choices.



Observability precedes controlled transition.

Postures are guideposts in a continuous space

Trust locates expected usefulness; Sovereignty locates retained judgement.



READING RULE

Guideposts, not identities

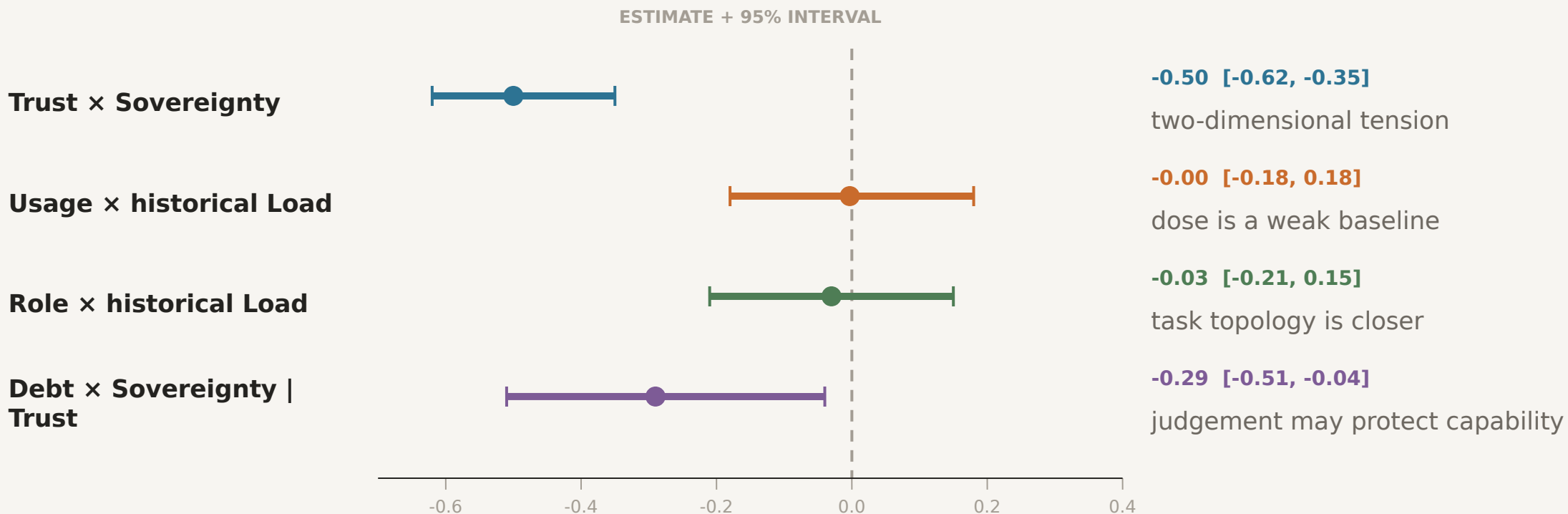
The same person can move by task, stakes, expertise, mandate and evidence.

Business use

**Route the coordination problem
—then test the protocol.**

Four signals point from dose toward structure

Associations become directional priors—and each prior has a decisive next test.



RESEARCH MOVE Measure task structure → randomize authorship and verification → validate against behaviour.

Different postures imply different business protocols

Four coordination opportunities; four minimal forms of selective friction.

1

PARTNERSHIP

Moving frontier

Light traceability: criteria, bounded delegation, sample checks

METRIC: REPAIR RISK

2

HYPERVIGILANCE

Verification tax

Risk tier + critical-edge tests + stopping rule

METRIC: DETECTION / CHECKING

3

OVERRELIANCE

Capability liability

Prior judgement + own-words explanation + delayed transfer

METRIC: RECONSTRUCTABILITY

4

COMPLIANCE

Displaced responsibility

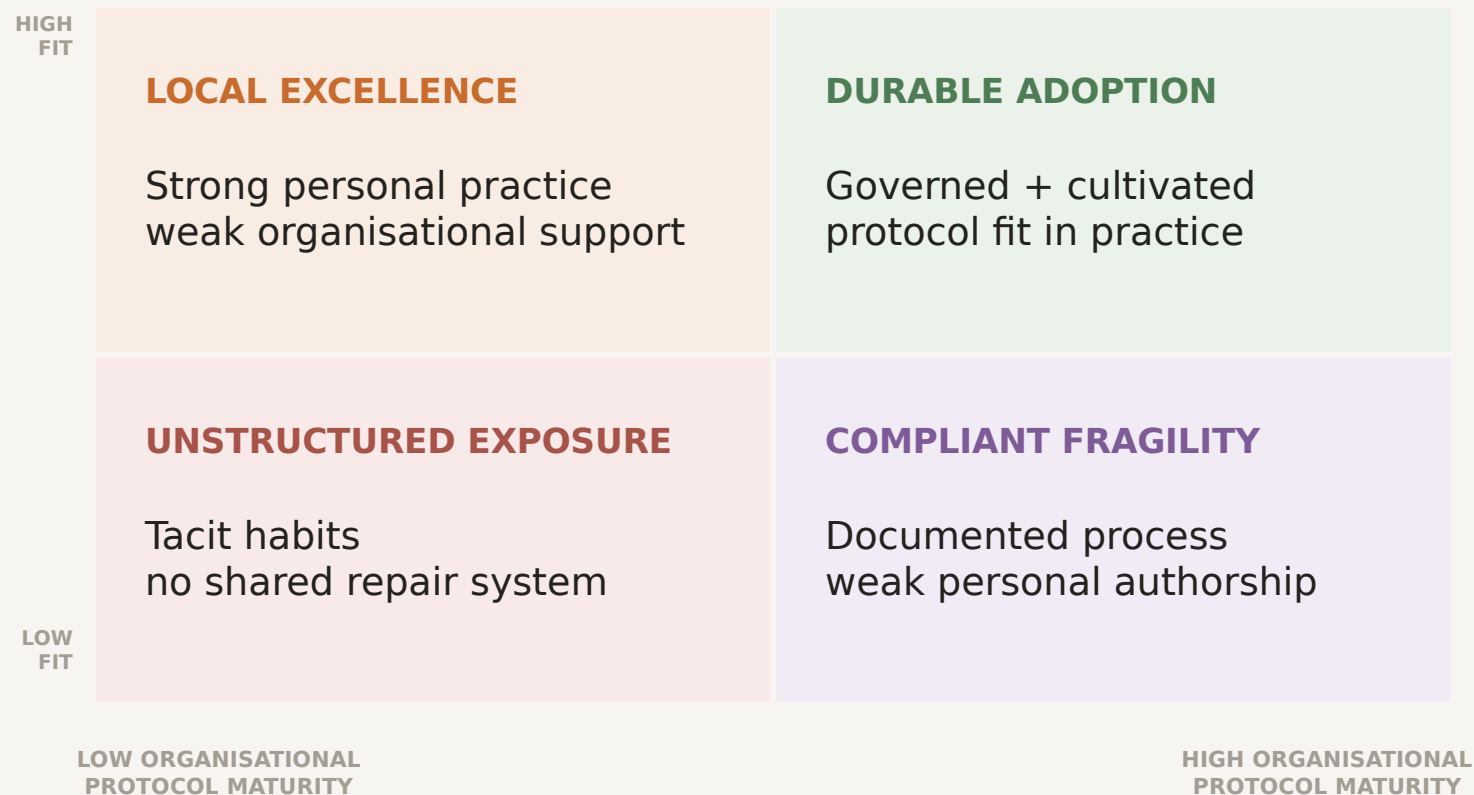
Named owner + sender gate + recourse

METRIC: RECIPIENT REWORK

Protocol fit = put ceremony where it buys learning, assurance or accountability.

Organisational maturity and individual posture must compose

Policy can look mature while lived coordination remains fragile—and vice versa.



SCALE CORRESPONDENCE

Disorientation → **DEBT**

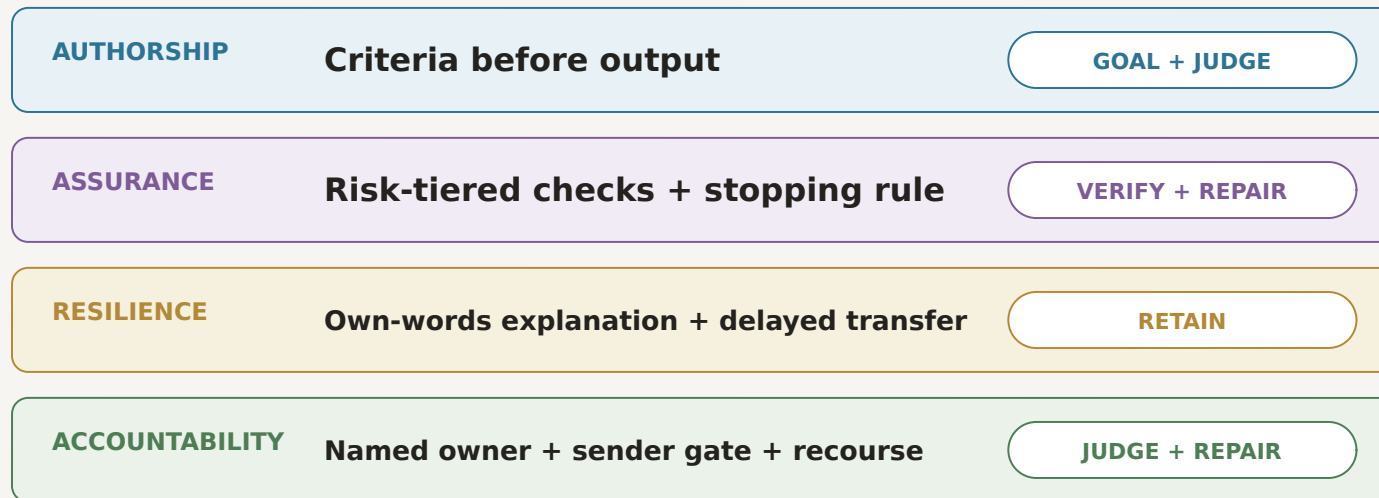
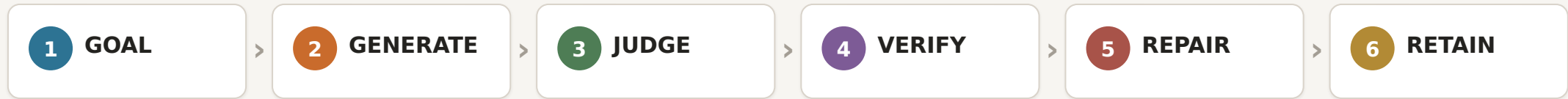
Tempo mismatch → **LOAD**

Extended blast radius → **SOV.**

The gap between policy and lived posture is an operational risk signal.

Protocol design is selective friction

Place ceremony where it produces learning, assurance or accountability.



VERSIONED PROTOCOL REGISTRY

- preconditions
- target failure mode
- required evidence
- expected transition
- maturity + provenance

Kits are testable compositions. The allocation problem is the theory.

Does protocol literacy transfer inward?

Commit the reading before the result: culture and governance remain complementary.

[PI COMMUNITY COHORT]

Responses close 7 September

Community vs prior pooled cohorts
on continuous Trust, Sovereignty, Load and Debt
with uncertainty intervals and n stated

LIVE REVEAL LANDING ZONE



HIGHER SOVEREIGNTY

Literacy may transfer; identify the mediating practices.



INDISTINGUISHABLE

Expertise may stay abstract; embed protocol in workflow.



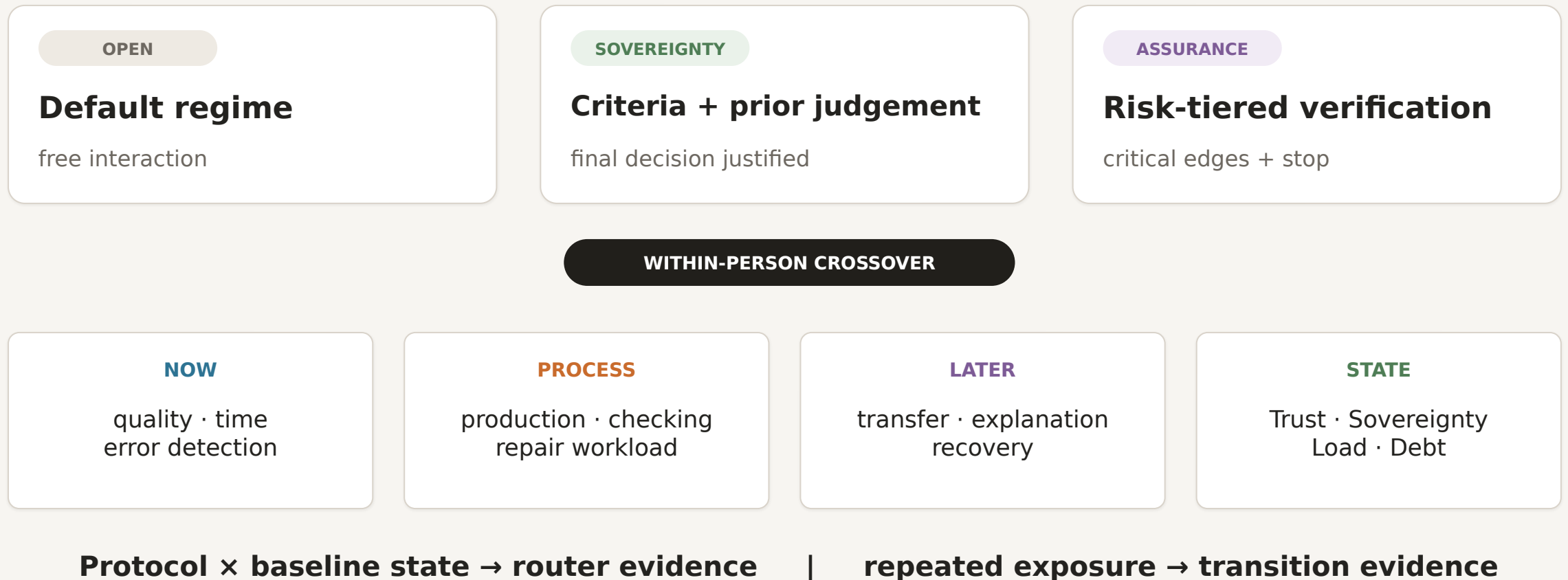
N TOO THIN

Retain direction; make recruitment part of infrastructure.

If literacy transfers: cultivate it. If it does not: encode it.

The first decisive experiment treats protocol as the intervention

Hold model, access, interface and task family constant; randomise the interaction rules.



One programme, two sets of deliverables

Formal representation and operating infrastructure advance together.

FOR PROTOCOL THEORY

STRUCTURE	roles · messages · traces · composition
SEMANTICS + LEGITIMACY	authority · responsibility · recourse
DYNAMICS	updates · transitions · co-evolution
VALUATION	output · capability · externalities · resilience

FOR BUSINESS

PAIRED DIAGNOSTIC	maturity × lived posture
PROTOCOL REGISTRY	versioned primitives + provenance
HANDOFF GATES	criteria · evidence · owner · escalation
OUTCOME LEDGER	cycle time · rework · traceability · recovery

SHARED INFRASTRUCTURE · evidence governance as a meta-protocol



The next layer of AI adoption is protocolization.

Make the interaction regime legible. Route protocols by posture. Measure value now and capability later.

1 FORMALISE

**protocol primitives
and composition**

2 RUN

**maturity × posture
field studies**

3 BUILD

**a versioned protocol
and evidence registry**

Every human-AI relationship already runs on a protocol. Can we read it, improve it, and learn from the transition?

Terms, precisely

INTERACTION REGIME

Recurring allocation of cognition and decision rights across person, model and organisation.

PROTOCOL

Conditional rule-set governing entry, roles, evidence, handoffs, repair and retention.

LOAD

Effort against capacity over time; current score isolates extraneous burden.

GROUNDING

Operational common ground without symmetrical commitment or accountability.

POSTURE

Provisional human-side state estimate of that regime; continuous and context-sensitive.

SOVEREIGNTY

Meaningful authorship of goals, criteria and consequential judgement.

COGNITIVE DEBT

Future capability liability when workings, skill or rationale go unfollowed.

EFFECTIVENESS

Immediate coupled performance plus distribution of cost and future capability.

BIDIRECTIONAL WITHIN THE EPISODE · ASYMMETRIC IN STAKES, PERSISTENCE AND ACCOUNTABILITY

Evidence register

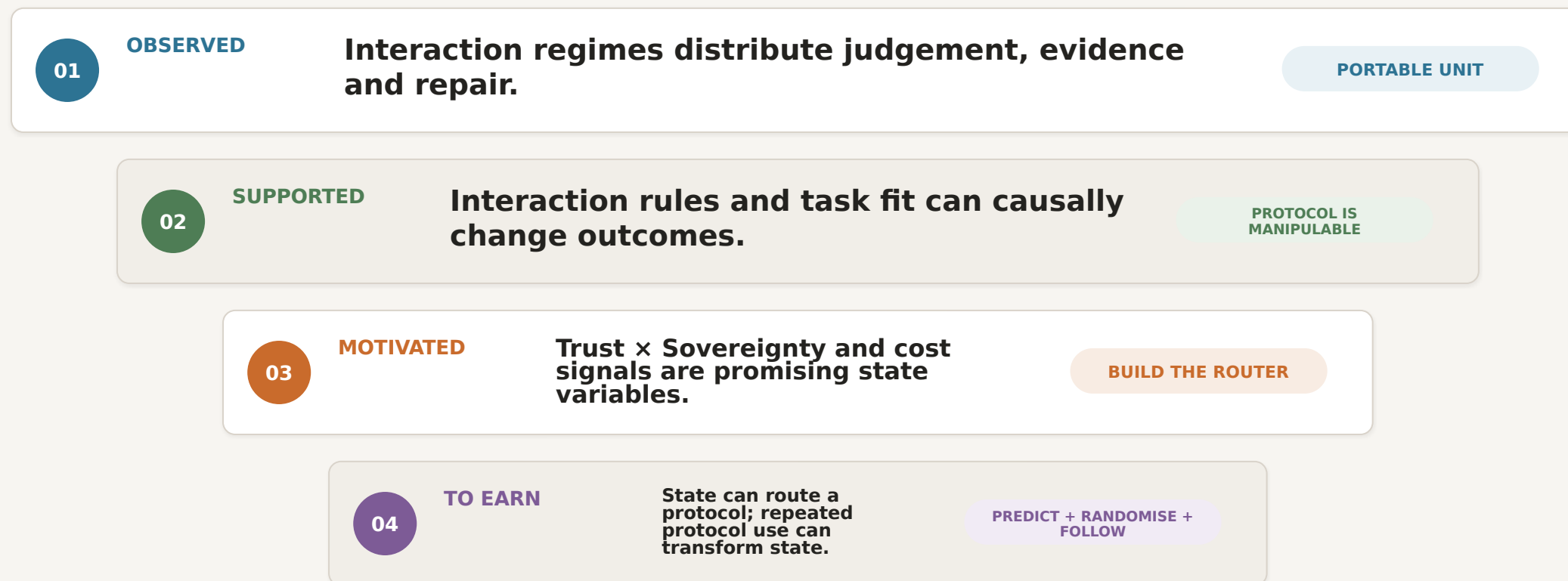
External causal anchors motivate the intervention level; internal signals shape the programme.

SOURCE	SCOPE	RESULT USED HERE	ROLE
Vaccaro et al.	106 experiments	augmentation $g = 0.64$; synergy vs best-alone $g = -0.23$	COORDINATION
Dell'Acqua et al.	758 consultants	inside-frontier gains; outside-frontier correctness fell	CONDITIONALITY
Bastani et al.	~1,000 students	same model; protocol changed assisted and later unaided outcomes	RETENTION
Internal pooled	$n = 121$	Trust $\times$ Sovereignty $r = -0.50$; dose and role weak Load proxies	STATE MAP
Internal sample	$n = 61$	Debt $\times$ Sovereignty Trust partial $r = -0.29$	MECHANISM

Evidence is layered by design: causal anchors → directional priors → prospective routing and transition tests.

Claims we can make today

A positive claim ladder: each level unlocks the next research and business decision.



Four failure modes, one person down

A bridge between Durable AI Adoption and the human-side state model.

ORGANISATIONAL	FAILURE MODE	INDIVIDUAL-SCALE EXPRESSION	SIGNAL
DISORIENTATION	Production outruns shared understanding	Workings outrun personal understanding	DEBT
TEMPO MISMATCH	Output stacks at the slowest verifier	Verification becomes the personal bottleneck	LOAD
EXTENDED BLAST RADIUS	Consequences spread beyond delegation	Judgement is ceded while accountability remains	SOVEREIGNTY
PREMATURE OPTIMISATION	Practice is standardised before it earns it	Study standardisation timing at organisation scale	OPEN

Measure both layers. The gap between them is itself evidence about protocol composition.

A versioned instrument is research infrastructure

Construct development becomes legible when changes, claims and score versions travel together.

CURRENT FROZEN ARCHITECTURE

33

unique items

3

scored
constructs

2D

Trust x
Sovereignty

3

verification
items

LOAD = EXTRANEOUS POLE

GERMANE EFFORT = RESEARCH

CHANGE LEDGER

- **LOAD SPLIT** productive effort separated from wasted burden
- **DISS RETIRED** no current score; mechanism becomes a fresh programme
- **CONTINUOUS FIRST** coordinates carry analysis; regions guide interpretation
- **CLAIMS VERSIONED** pre-registration, register and changelog travel together

THE RESEARCH PACKAGE IS ITSELF AN EPISTEMIC PROTOCOL